# LAB Assignment No 9

# Distance-Based LED Control using Ultrasonic Sensor with ESP32 and ThingSpeak

In this lab, students will develop an IoT-based distance monitoring system using an ESP32 and an ultrasonic sensor. The system measures the distance of an object and classifies it into three ranges. Based on the detected distance, different LEDs will be activated. The measured distance will also be sent to the cloud using ThingSpeak for real-time visualization and monitoring.

**Lab Objectives**

- To interface **Ultrasonic Sensor (HC-SR04)** with ESP32
- To measure distance of an object
- To implement multi-level LED control logic
- To send distance data to ThingSpeak cloud
- To visualize distance data using cloud graph

**Learning Outcomes**

After completing this lab, students will be able to:

- Understand working of ultrasonic sensors
- Measure distance using ESP32
- Implement multi-condition logic (if-else)
- Control multiple LEDs based on sensor data
- Send IoT data to cloud platform
- Visualize and analyze distance data

**Required Hardware / Software**

**Hardware**

- ESP32 Development Board
- Ultrasonic Sensor (HC-SR04)
- 3 LEDs (LED1, LED2, LED3)
- 3 × 220Ω Resistors

- Breadboard & Jumper Wires

**Software**

- Arduino IDE
- Wokwi Simulator
- ThingSpeak Account

## 4. Circuit Diagram

### 🔌 Connections:

| Component | ESP32 Pin |
|---|---|
| VCC (Ultrasonic) | 5V |
| GND | GND |
| TRIG | GPIO 5 |
| ECHO | GPIO 18 |
| LED1 | GPIO 16 |
| LED2 | GPIO 17 |
| LED3 | GPIO 19 |

## 🔷 5. Working Principle

1. Ultrasonic sensor sends sound waves
2. Measures echo time to calculate distance
3. ESP32 processes distance and applies logic:

| Distance Range | LEDs ON |
|---|---|
| < 10 cm | LED1 |
| 10–15 cm | LED1 + LED2 |
| > 15 cm | LED1 + LED2 + LED3 |

4. Distance value is displayed on Serial Monitor
5. Data is sent to ThingSpeak every 15 seconds

**Program Code**

```
#include <WiFi.h>
#include <ThingSpeak.h>

// WiFi credentials
const char* ssid = "Wokwi-GUEST";
const char* password = "";

// ThingSpeak
unsigned long channelID = YOUR_CHANNEL_ID;
const char* writeAPIKey = "YOUR_API_KEY";

WiFiClient client;

// Ultrasonic pins
#define TRIG_PIN 5
#define ECHO_PIN 18

// LED pins
#define LED1 16
#define LED2 17
#define LED3 19

void setup() {
```

```
  Serial.begin(115200);

  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);

  pinMode(LED1, OUTPUT);
  pinMode(LED2, OUTPUT);
  pinMode(LED3, OUTPUT);

  WiFi.begin(ssid, password);
  Serial.print("Connecting");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nConnected!");

  ThingSpeak.begin(client);
}

void loop() {
  // Trigger ultrasonic pulse
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  long duration = pulseIn(ECHO_PIN, HIGH);

  // Calculate distance (cm)
  float distance = duration * 0.034 / 2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  // LED Logic
  if (distance < 10) {
    digitalWrite(LED1, HIGH);
    digitalWrite(LED2, LOW);
    digitalWrite(LED3, LOW);
  }
  else if (distance >= 10 && distance <= 15) {
```

```
    digitalWrite(LED1, HIGH);
    digitalWrite(LED2, HIGH);
    digitalWrite(LED3, LOW);
  }
  else {
    digitalWrite(LED1, HIGH);
    digitalWrite(LED2, HIGH);
    digitalWrite(LED3, HIGH);
  }

  // Send data to ThingSpeak
  ThingSpeak.setField(1, distance);

  int response = ThingSpeak.writeFields(channelID, writeAPIKey);

  if (response == 200) {
    Serial.println("Data sent to ThingSpeak");
  } else {
    Serial.print("Error: ");
    Serial.println(response);
  }

  delay(15000); // ThingSpeak delay
}
```

**ThingSpeak Setup**

- Create a channel
- Add:
  - Field 1 → Distance
- Copy:
  - Channel ID
  - Write API Key

**Expected Outputs**

 Simulation (Wokwi)

- Distance changing dynamically
- LEDs turning ON based on distance

- Serial Monitor showing distance

📊 Cloud Output (ThingSpeak)

- Distance graph
- Real-time updates

## Lab Tasks

Students must:

✅ Interface ultrasonic sensor with ESP32
✅ Implement 3-level LED control
✅ Display distance on serial monitor
✅ Send data to ThingSpeak
✅ Observe and analyze cloud graph

## Deliverables

- Source code
- Wokwi simulation screenshot
- ThingSpeak graph screenshot
- Brief report

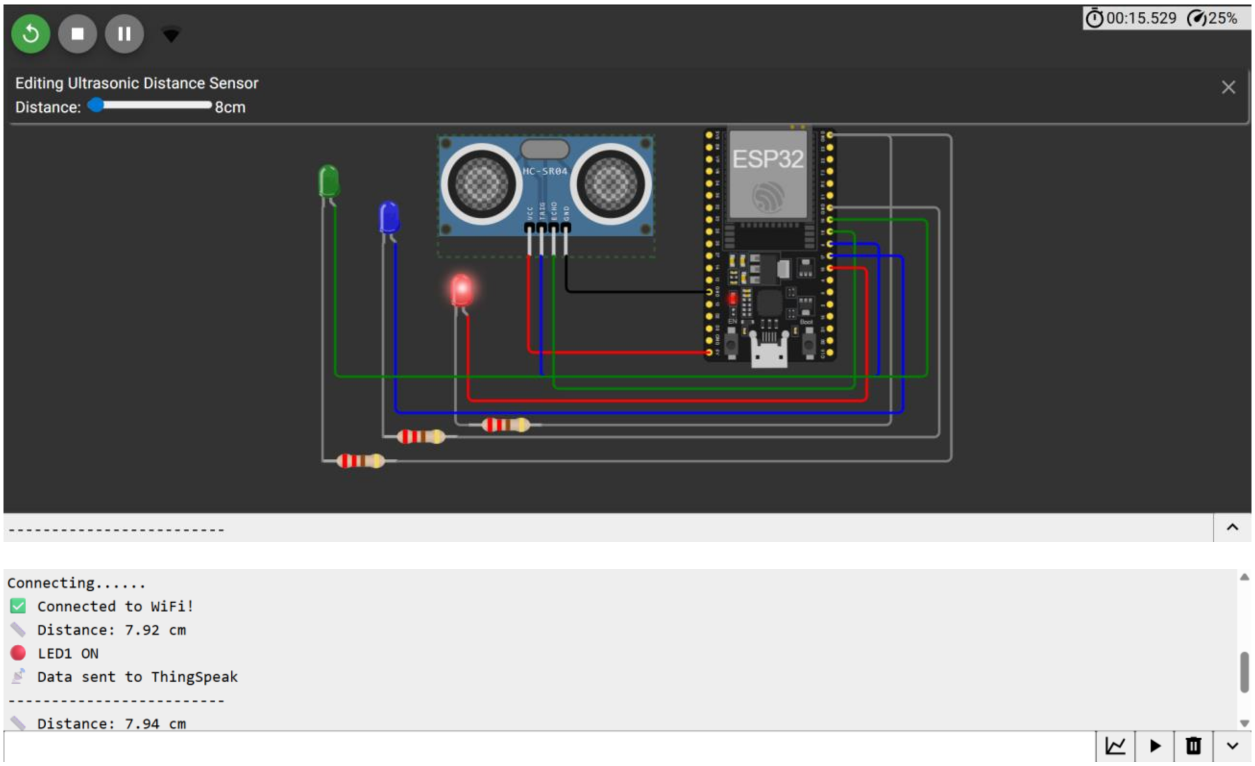
**Viva Questions**

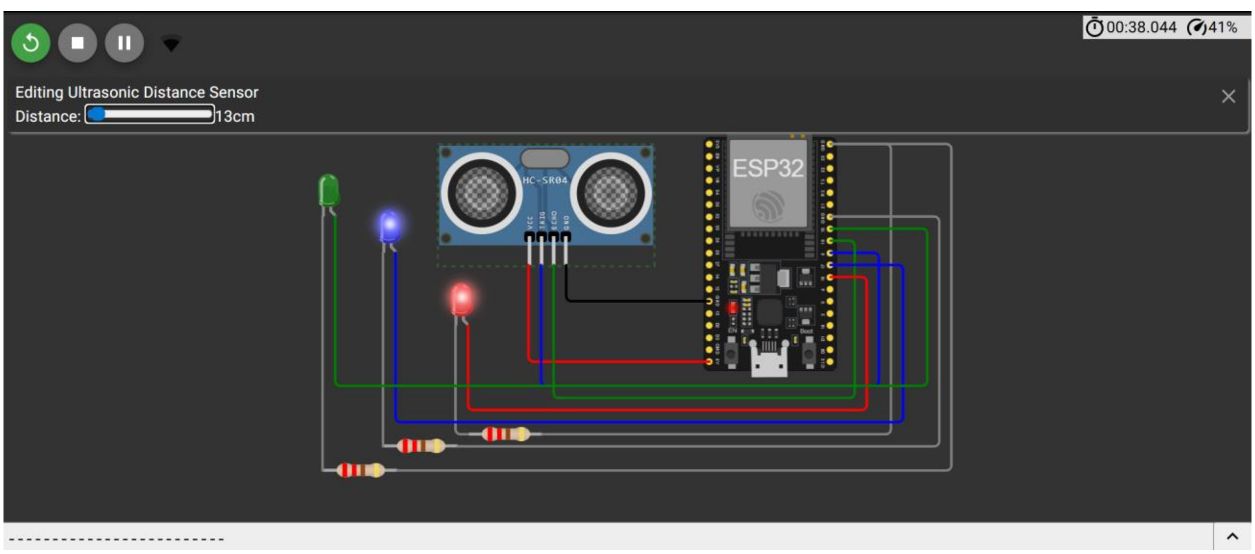
1. What is ultrasonic sensor?
2. How distance is calculated?
3. What is speed of sound used?
4. Why divide by 2 in formula?
5. What is ThingSpeak?
6. Why delay is required?

```cpp
#include <WiFi.h>
#include <ThingSpeak.h>

// WiFi credentials
const char* ssid = "Wokwi-GUEST";
const char* password = "";

// ✅ Your ThingSpeak details
unsigned long channelID = 3366883;
const char* writeAPIKey = "RPN46HCX0M0391KK";

WiFiClient client;

// Ultrasonic pins
#define TRIG_PIN 5
#define ECHO_PIN 18

// LED pins
#define LED1 16
#define LED2 17
#define LED3 19

void setup() {
  Serial.begin(115200);

  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);

  pinMode(LED1, OUTPUT);
  pinMode(LED2, OUTPUT);
  pinMode(LED3, OUTPUT);

  WiFi.begin(ssid, password);
  Serial.print("Connecting");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\n✅ Connected to WiFi!");

  ThingSpeak.begin(client);
}

void loop() {
  // Trigger ultrasonic pulse
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
```

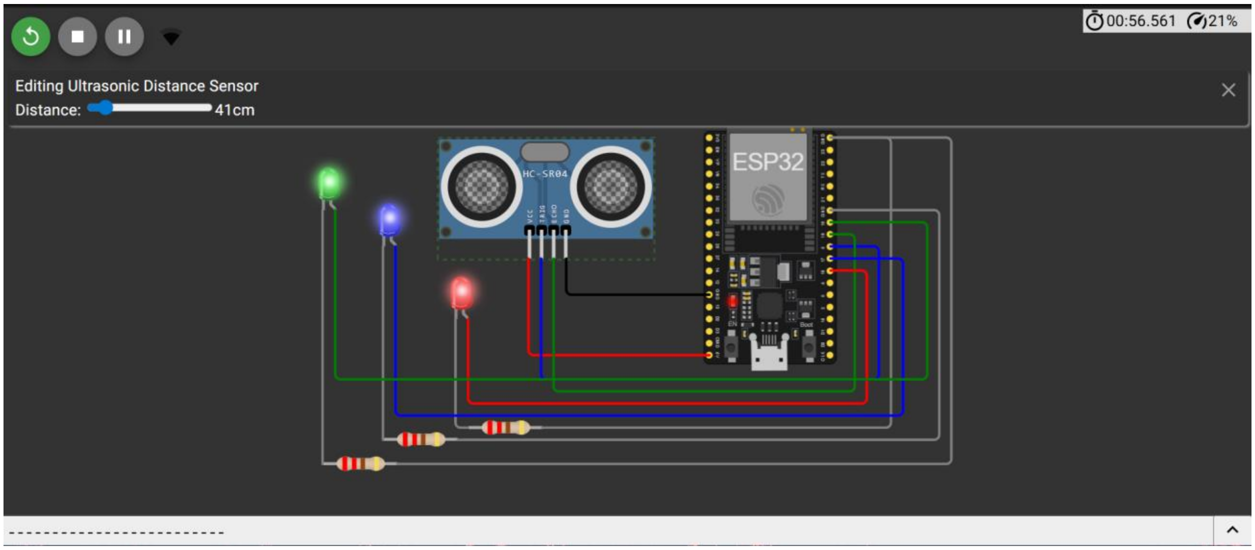
```
digitalWrite(TRIG_PIN, HIGH);
delayMicroseconds(10);
digitalWrite(TRIG_PIN, LOW);

long duration = pulseIn(ECHO_PIN, HIGH);

// Calculate distance (cm)
float distance = duration * 0.034 / 2;

Serial.print("📏 Distance: ");
Serial.print(distance);
Serial.println(" cm");

// LED Logic
if (distance < 10) {
  digitalWrite(LED1, HIGH);
  digitalWrite(LED2, LOW);
  digitalWrite(LED3, LOW);
  Serial.println("🔴 LED1 ON");
}
else if (distance >= 10 && distance <= 15) {
  digitalWrite(LED1, HIGH);
  digitalWrite(LED2, HIGH);
  digitalWrite(LED3, LOW);
  Serial.println("🟡 LED1 + LED2 ON");
}
else {
  digitalWrite(LED1, HIGH);
  digitalWrite(LED2, HIGH);
  digitalWrite(LED3, HIGH);
  Serial.println("🟢 ALL LEDs ON");
}

// Send data to ThingSpeak
ThingSpeak.setField(1, distance);

int response = ThingSpeak.writeFields(channelID, writeAPIKey);

if (response == 200) {
  Serial.println("📡 Data sent to ThingSpeak");
} else {
  Serial.print("❌ Error: ");
  Serial.println(response);
}

Serial.println("-------------------------");

```

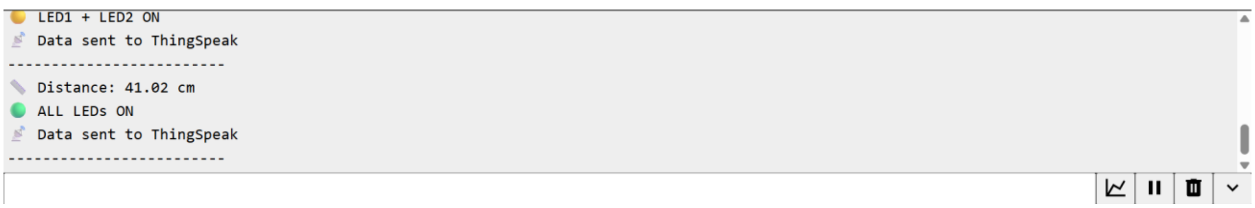
```
  Serial.println("------------------------");

  delay(15000); // ThingSpeak delay
}
```

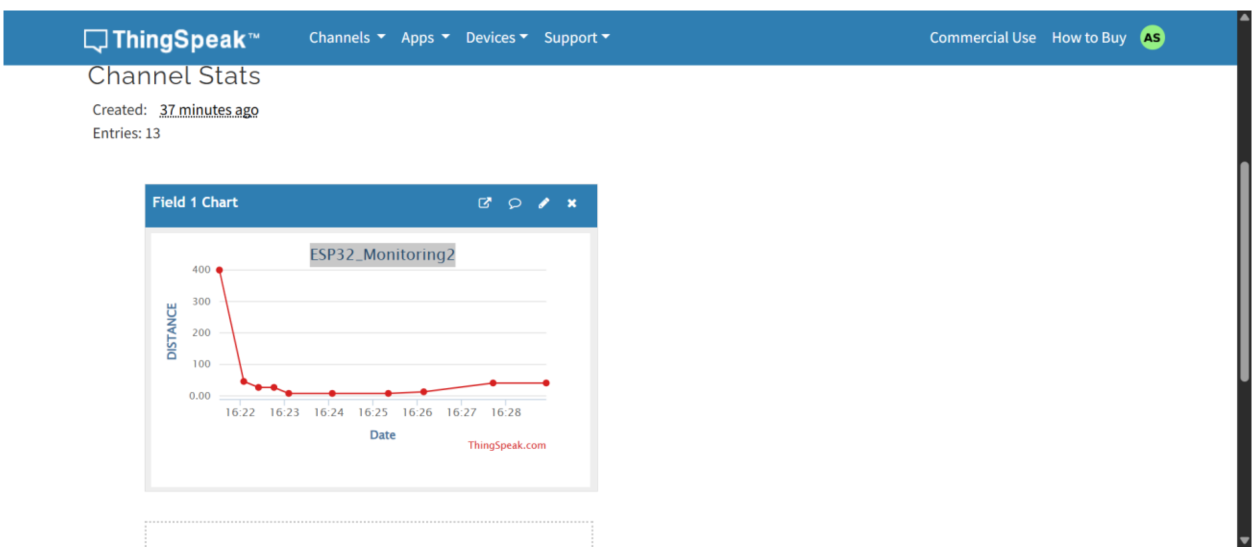



Distance: 7.94 cm
LED1 ON
Data sent to ThingSpeak
------------------------
Distance: 13.02 cm
LED1 + LED2 ON
Data sent to ThingSpeak


00:56.561
21%
Editing Ultrasonic Distance Sensor
Distance:
41cm
HC-SR04
ESP32
------------------------


LED1 + LED2 ON
Data sent to ThingSpeak
------------------------
Distance: 41.02 cm
ALL LEDs ON
Data sent to ThingSpeak
------------------------


ThingSpeak™
Channels
Apps
Devices
Support
Commercial Use
How to Buy
AS
Channel Stats
Created: 37 minutes ago
Entries: 13
Field 1 Chart
ESP32_Monitoring2
DISTANCE
400
300
200
100
0.00
16:22
16:23
16:24
16:25
16:26
16:27
16:28
Date
ThingSpeak.com

## LAB Assessment

| | | | |
|---|---|---|---|
| **Student Name** | | **LAB Rubrics** | CLO3, P5, PLO5 |
| | | **Total Marks** | 10 |
| **Registration No** | | **Obtained Marks** | |
| | | **Teacher Name** | Dr. Syed M Hamedoon |
| **Date** | | **Signature** | |